Bayesian Learning & MonteCarlo Simulation Final Project

Hourly Load Factor

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Description:

The project explores the application of Bayesian Learning techniques and MonteCarlo simulations to analyze and predict the power usage of the city of Tetouan, Morocco.

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July 2026

1 Introduction

1.1 Data Description

This report analyzes the hourly power consumption in the city of Tetouan, Morocco, during 2017. We use a pre-cleaned sample of 2,000 observations representing the hourly load factor:

$$LF(d,h) = \frac{P(d,h)}{\max_{h'} P(d,h')}$$

where $P(d,h)$ is the total active power consumed across all city zones on day d and hour h. The target variable is bounded, $LF \in [0,1]$, and has been clamped to $(10^{-3}, 1-10^{-3})$ to ensure finite Beta log-densities.

The used dataset is structured as:

Date	**Hour**	**LF**

1.2 Objectives

As the load factor is strictly bounded, we model it using Beta distributions. The analysis proceeds in four steps:

1. **Model Selection:** Fit a single Beta model and Beta mixtures with 1 to 5 components, using WAIC for selection.
2. **Posterior Checking:** Overlay the posterior predictive density on the empirical histogram to verify mode recovery.
3. **Post-Hoc Exploration:** Assign observations to their Maximum A Posteriori (MAP) components and plot the component proportions by hour.
4. **Temporal Modeling:** Fit a multinomial logistic regression predicting the MAP component assignments. We use Fourier-transformed hours (sine and cosine terms) as covariates to capture daily periodicity, and compare predicted probabilities with the empirical post-hoc plots.

2 Task 1: Model Selection

2.1 Model Specification

As requested by the task we aim to select an appropriate model for our dataset, this is requested to be a Beta distribution, which is an adequate fit since the load factor is strictly bounded: $LF \in (0,1)$. More specifically we consider both a single Beta model and a mixture of $H = 2, 3, 4, 5$ Beta components to capture the multimodal nature of daily power consumption.

For the single Beta model ($H = 1$), the likelihood is specified as

$$y_i \sim \text{Beta}(\alpha, \beta)$$

We assign weakly informative priors to the shape parameters: $\alpha, \beta \sim \text{Gamma}(0.1, 0.1)$. This flat, heavy-tailed prior ensures the parameters remain positive while allowing the data to drive the posterior.

For the mixture models ($H > 1$), the data is assumed to be generated from H distinct components. The likelihood is given by:

$$p(y_i \mid \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\pi}) = \sum_{h=1}^{H} \pi_h \text{Beta}(y_i \mid \alpha_h, \beta_h)$$

where $\boldsymbol{\pi}$ represents the assigned weights to each distribution.

This mixture is implemented practically by introducing a latent variable, z_i, for each observation y_i, which indicates the specific component from which the observation is drawn. z_i follows a categorical distribution parameterized as:

$$z_i \sim \text{Categorical}(\boldsymbol{\pi})$$

To formulate the priors and prevent JAGS from label switching, we reparameterize the Beta components in terms of their mean μ_h and concentration κ_h:

$$\mu_h = \frac{\alpha_h}{\alpha_h + \beta_h}, \quad \kappa_h = \alpha_h + \beta_h$$

This allows us to recover the shape parameters as

$$\alpha_h = \mu_h \kappa_h, \quad \beta_h = (1 - \mu_h)\kappa_h$$

.

Regarding the priors, we applied these choices:

- The means are treated with uninformed priors as uniform distributions, $\mu_h \sim \text{Uniform}(0, 1)$, reflecting no initial knowledge on where the consumption modes are located.
- The concentrations are given weakly informative priors, $\kappa_h \sim \text{Gamma}(0.1, 0.1)$.
- The mixing weights are assigned a symmetric, uninformative Dirichlet prior, $\boldsymbol{\pi} \sim \text{Dirichlet}(1, 1, \ldots, 1)$ to signal unknown significance of each component in the complete distribution.

As mentioned earlier we use means instead of a direct α, β approach to prevent the MCMC sampler from label switching. This issue presents itself with mixture models in JAGS, to avoid it we impose a strict ordering constraint on the means such that $\mu_1 < \mu_2 < \cdots < \mu_H$, this implies the labels are always ordered and correctly assigned.

2.2 Posterior Analysis and MCMC Algorithm

To approximate the posterior distributions, we implemented Markov Chain Monte Carlo (MCMC) methods using JAGS (Just Another Gibbs Sampler), which relies on Gibbs sampling to explore the parameter space. For each model, we ran 2 parallel chains. Each chain was subjected to a burn-in phase of 700 iterations to allow the sampler to find the high-probability regions of the posterior, as for the inference we ran 3 different scenarios with 1000, 2000, and 3000 iterations to obtain a better view of how changing the execution time would impact the results.

2.3 Model Selection

To perform a valid model selection (choosing the most appropriate mixture of Beta distributions), we compared the predictive accuracy of the models using the Watanabe-Akaike Information Criterion (WAIC).

To compute the WAIC, we evaluate the marginal log-likelihood for each observation y_i. By marginalizing over the unknown latent assignments z_i, we weight the likelihood of each component by its mixing proportion π_h:

$$\log \left(\sum_{h=1}^{H} \pi_h \cdot \text{Beta}(y_i \mid \alpha_h, \beta_h) \right)$$

This accounts for our uncertainty regarding latent assignments, ensuring an unbiased evaluation of the model's overall predictive performance. The model with the lowest WAIC score is selected as the optimal representation of the data, balancing goodness-of-fit against model complexity.

The results we obtained across the different executions are visible in this table:

Model	1000 Iterations	2000 Iterations	3000 Iterations
H = 1	-1634.36	-1633.79	-1634.11
H = 2	-2689.84	-2689.47	-2490.29
H = 3	-2800.12	-2988.61	-3098.91
H = 4	-2810.93	-3114.57	-3105.56
H = 5	-3107.15	-3110.98	-3108.55

Table 1: WAIC values across different model complexities and MCMC iteration settings.

We can see that by changing the parameters the best model varies between a mixture of $H = 4$ and $H = 5$. The chosen models have a very small difference in WAIC, implying there shouldn't be any major difference between the different model complexities. We can better visualize the distinction between the two mixtures in the following chapter.

3 Task 2: Posterior Checking

3.1 Posterior Predictive Checking and Mode Recovery

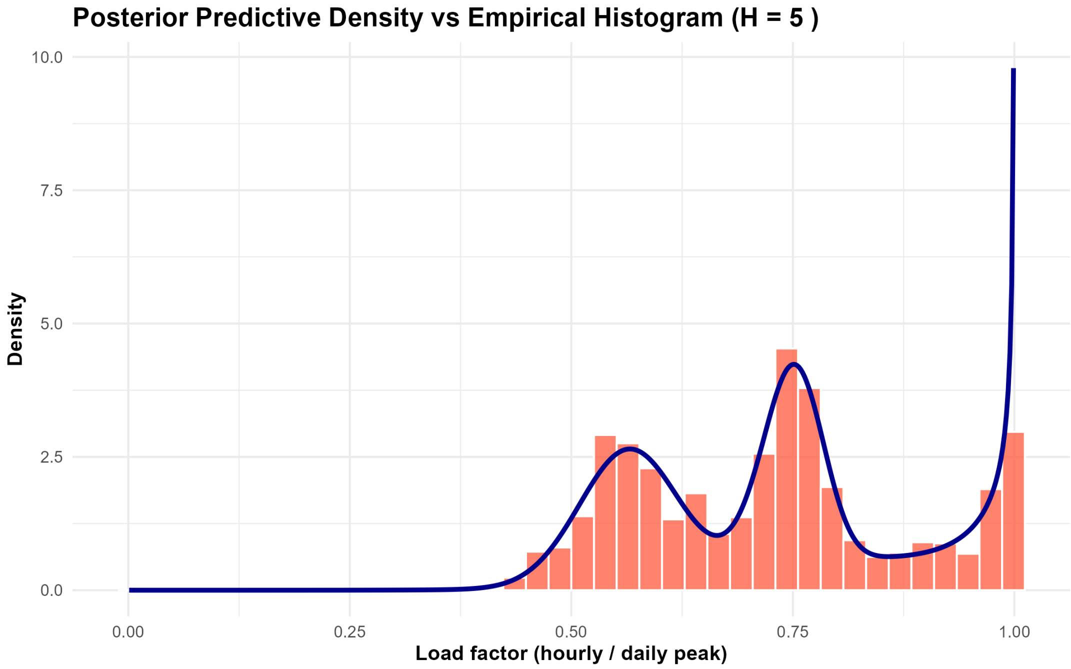
The second task requires validating our chosen model, and while the WAIC balances fit and complexity, we wish to visually verify whether the theoretical data distribution generated by our model accurately reflects the actual power consumption patterns of Tetouan. Specifically, we aim to confirm that the model successfully recovers the different visible peaks (modes).

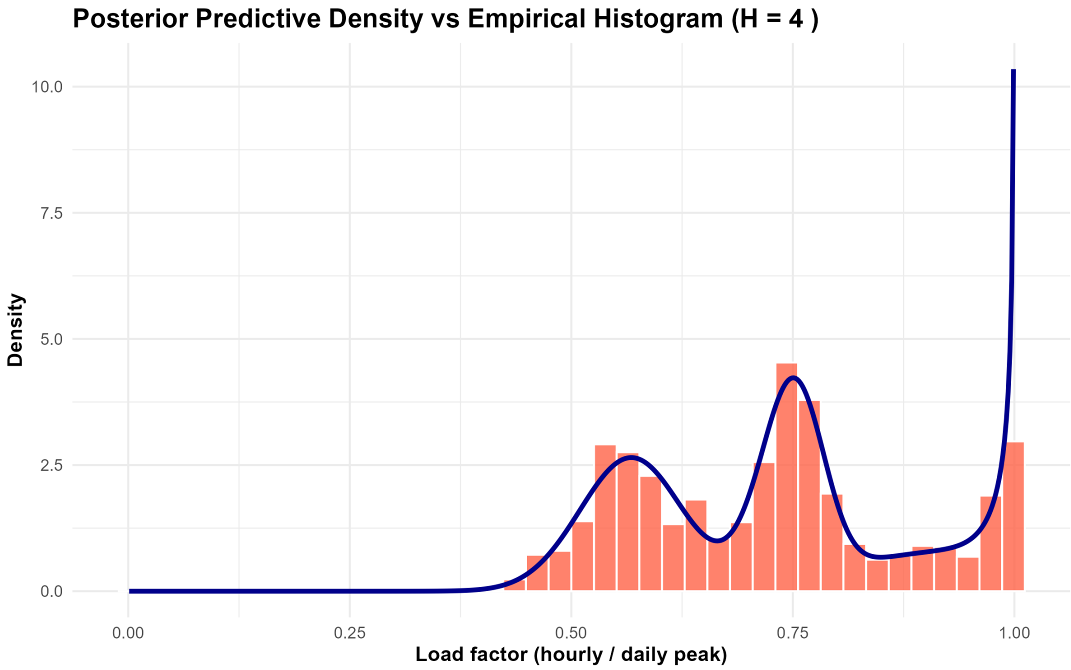
To achieve this, we calculate the Bayesian posterior predictive density. Rather than relying on a single point estimate (such as the mean of the parameters), we marginalize over the uncertainty by utilizing the full set of MCMC samples generated previously. For a grid of hypothetical future observations $\tilde{y}$, the expected predictive density is approximated by averaging over the S MCMC draws:

$$p(\tilde{y} \mid y) \approx \frac{1}{S} \sum_{s=1}^{S} \sum_{h=1}^{H} \pi_h^{(s)} \text{Beta}(\tilde{y} \mid \alpha_h^{(s)}, \beta_h^{(s)})$$

By computing this weighted sum of Beta distributions for every single MCMC step and averaging the results, we fully propagate the parameter uncertainty into our final curve.

We can visualize this averaged theoretical probability curve overlaid onto the empirical histogram of the load factor data:

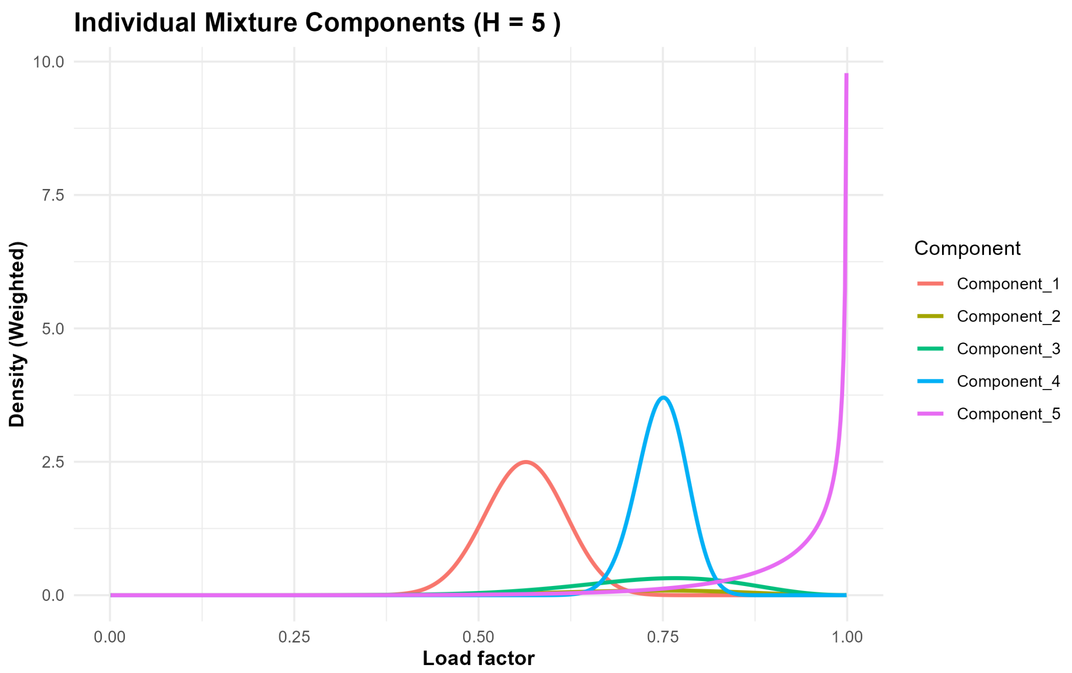


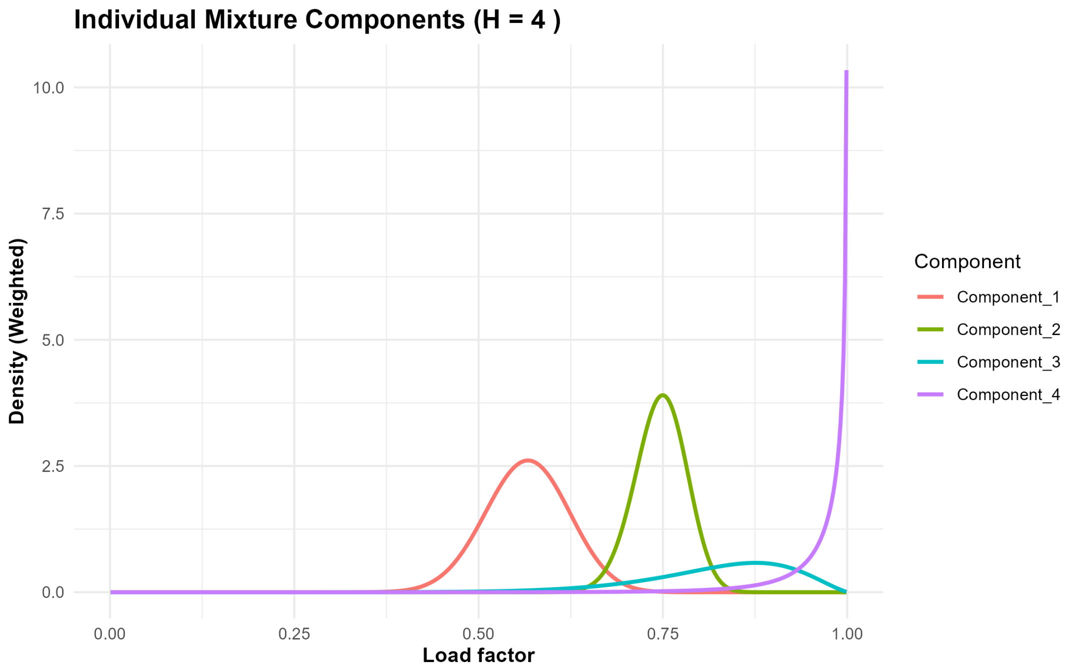


As we can see the modes were recovered in an appropriate way by both models, and the resulting distributions are almost identical, verifying the small difference presented between the WAIC values of these two mixture models and the constant switching between the best model choice.

3.2 Sub-Component Analysis

To better understand the components and mechanics of the mixture model we also extract and plot the individual weighted mixture components, this analysis reveals how the algorithm has partitioned the data. Each distinct Beta curve effectively handles a specific consumption regime, visualizing the sub-populations that construct the overall power grid demand.





Analyzing the dissected structures we can finally understand the underlying difference between the two models. We can see how the two models distinguish themselves, noticing that the $H = 5$ model introduces an almost flat beta distribution which has very low significance across the complete model, making the weighted sum of these components practically identical (as viewed before).

Seeing this extreme similarity we focus on the simpler model ($H = 4$) as a reference. We can better analyze and identify extracted sub-groups:

- Component 1: Base load of the city
- Component 2: Medium consumption of power

- Component 3: Very low peaked with a high spread, it represents a transitioning phase between peak an base loads

- Component 4: standing at a LF of 1 it represents the peak load of power consumption.

(These intuitions on the different clusters can be further proved and observed by the following tasks).
For the next stage of the Load Factor analysis we will only focus on the 4 component mixture model, seeing the general equivalence of the two (previously shown) and the ease of analysis of a lower complexity model.

4 Task 3: Post-Hoc Exploration

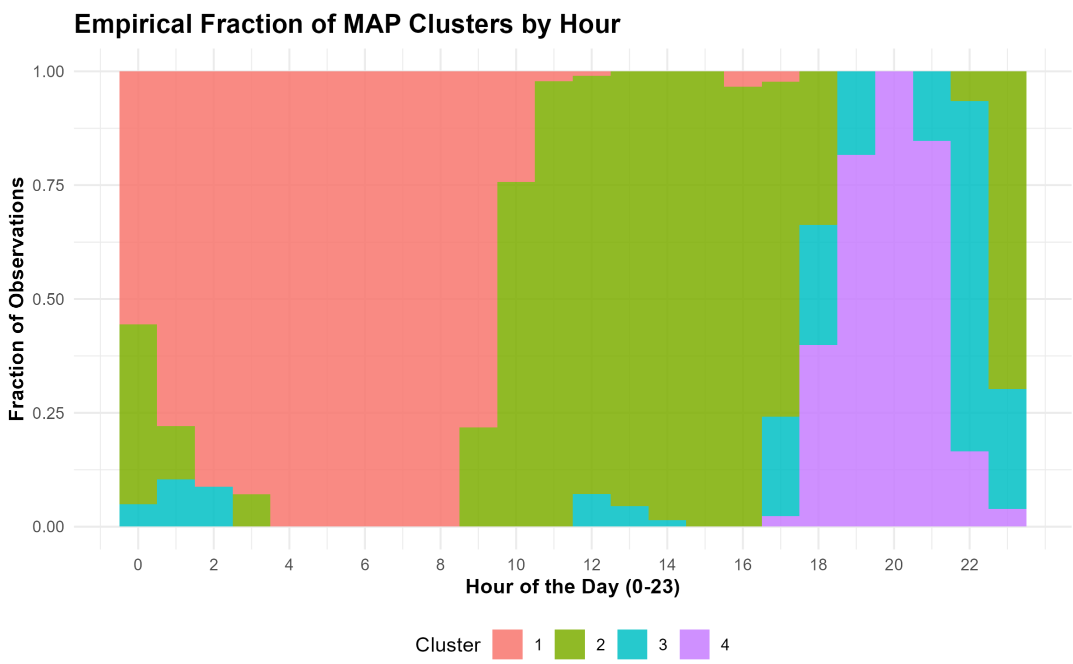
Although the Beta mixture model successfully captures the various power consumption regimes, it doesn't allow for a comprehensive view of how these regimes operate throughout the day. To understand the temporal dynamics, we performed a post-hoc exploration.

First, we mapped every observation y_i to its most likely underlying mixture component by calculating the Maximum A Posteriori (MAP) assignment. Using the posterior means of our fitted parameters $(\hat{\pi}_h, \hat{\alpha}_h, \hat{\beta}_h)$, we applied Bayes' theorem to compute the probability that observation i belongs to cluster h:

$$\hat{P}(Z_i = h \mid y_i) = \frac{\hat{\pi}_h \text{Beta}(y_i \mid \hat{\alpha}_h, \hat{\beta}_h)}{\sum_{k=1}^{H} \hat{\pi}_k \text{Beta}(y_i \mid \hat{\alpha}_k, \hat{\beta}_k)}$$

Each observation was then assigned to the cluster that maximized this probability.

Following the MAP assignment, we grouped the dataset by the hour of the day (from hour 0 to 23). Then for each hour, we calculated the empirical fraction of observations belonging to each component. The respecive fractions can be seen on the following graph:



We can see that the plot reveals a somewhat expected behavior of the power grid, with the base load occupying the nightly hours fading into an average load throughout the afternoon and peaking with the max load around the evening, with cluster 3 acting as a transition phase.

5 Task 4: Temporal Modeling

5.1 Advanced Temporal Modeling: Multinomial Logistic Regression

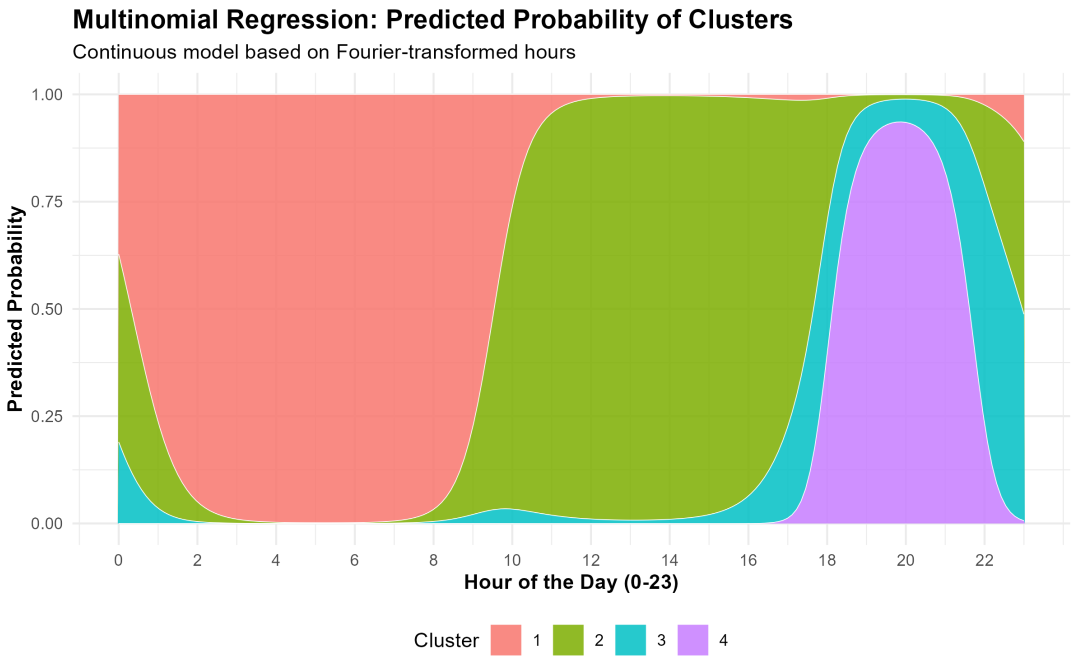
While the post-hoc exploration revealed the empirical distribution of the clusters throughout the day, we further aimed to formally model this temporal dependency. The final objective is to construct a predictive classification model allowing us to estimate the probability that the power consumption at a certain hour belongs to a specific cluster.

To do this, we used the MAP component assignments from the previous step as our discrete response variable. Since the output consists of H categories we fitted a multinomial logistic regression model.

To encode the information of hours looping in a day (24:00 being equal to 00:00) we apply Fourier transformations, converting the hour into a cyclical coordinate system using the fundamental 24-hour frequency. We also included the 12-hour harmonic to allow the model to capture secondary daily peaks, giving more complexity to the output. The predictor for cluster h is structured as:

$$\eta_h = \beta_{h,0} + \beta_{h,1} \sin\left(\frac{2\pi \text{hour}}{24}\right) + \beta_{h,2} \cos\left(\frac{2\pi \text{hour}}{24}\right) + \beta_{h,3} \sin\left(\frac{4\pi \text{hour}}{24}\right) + \beta_{h,4} \cos\left(\frac{4\pi \text{hour}}{24}\right)$$

Using the model, we predicted the cluster probabilities over the same 24-hour grid shown previously:



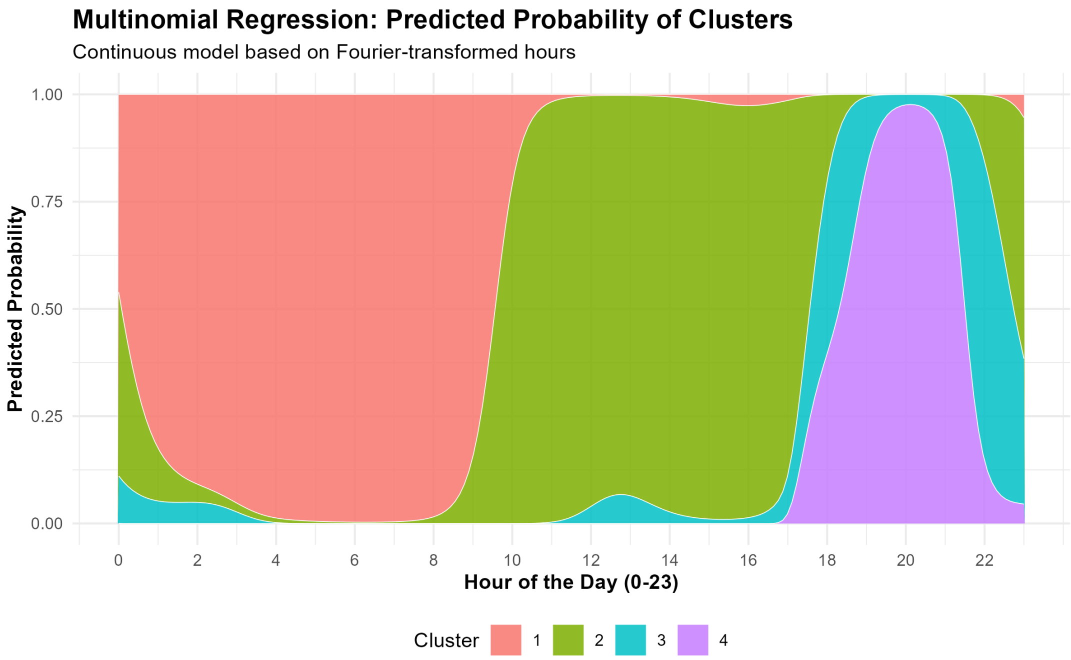
5.2 Results Comparison

Comparing the continuous predictions with the empirical MAP assignments we can see there is an evident alignment between the two. The continuous model smooths out the raw empirical counts while still preserving the transitions between the power regimes and their individual peaks. This demonstrates that the Fourier-based model is effective at predicting the cyclical pattern of Tetouan's power consumption.

5.3 Additional Harmonics Insertion

As previously stated, we introduced the 12-hour harmonic for more complex analysis by the model, on a theoretical level this approach could've been expanded to more complex models introducing the 8-hour, 6-hour, etc. harmonics.

Adding more and more features would allow for a more detailed analysis, but could also increase model complexity by a non-insignificant factor. To verify the need of additional harmonics we tested the model with the additional 8-hour harmonic, as we can see in the graph:



The graph shows that this addition was able to capture some extra details, such as the small peak of cluster 3 at 2 am, and some deformations of the curves, yet the differences were small and a simpler model could suffice to give a good prediction estimate.